THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** wonder through why and how questions	Science begins with wonder
2	**(a)** temperature of the oil	A relevant controllable factor
3	**(c)** making sense of the results later	Records are evidence
4	**(c)** nucleus	Genetic material lies in the nucleoid
5	**(d)** cow dung and organic waste	Anaerobic breakdown
6	**(b)** legumes	Peas, beans, lentils
7	**(c)** training the immune system	Preventive, not curative
8	**(a)** iron	Low haemoglobin
9	**(c)** Typhoid	Caused by bacteria, spreads
10	**(c)** resistance of the conductor	Electrical energy converts to heat
11	**(b)** moist paste	Not a liquid
12	**(c)** electrolyte	Liquid or paste

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	To know which change caused the effect (1). Other factors kept same keep the test fair (1).	1+1
14	Region holding genetic material without a membrane (1). Bacterial cells (1).	1+1
15	It lives in root nodules of legumes (1). It fixes atmospheric nitrogen and increases soil fertility (1).	1+1
16	Communicable diseases are caused by pathogens and spread from person to person (1). Non-communicable diseases are caused by lifestyle, diet or environment and do not spread (1).	1+1
17	A coil of wire that behaves like a magnet when current flows (1). Strengthened by an iron core (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Many factors act together (1). Scientists still investigate it (1). It shows science is never finished (1).	1+1+1
19	Cow dung and waste are put in a closed tank (1). Microbes break it down producing methane-rich gas (1). Gas is used for cooking and lighting (1).	1+1+1
20	Cause: pathogens vs lifestyle, diet or environment (1). Spread: yes vs no (1). Examples: typhoid; diabetes (1).	1+1+1
21	More current gives stronger field (1). More turns gives stronger magnet (1). Iron core makes it much stronger (1).	1+1+1
22	Magnitude of current (1). Material, thickness and length of wire (1). Duration of current (1).	1+1+1
	OR: Electric heater (1). Electric iron or kettle (1). Incandescent lamp (1).	1+1+1
23	Each answer reveals new unknowns (1½). Science keeps evolving as ideas are tested (1½).	1½+1½
	OR: Moon phases repeat in a cycle (1). Observing sunrise, sunset and Moon cycles gave calendars (1). So earthly routines link to the sky (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Shape suits job (1). Nerve cells are long and branched to carry signals (1½). Red blood cells are small discs to move easily in vessels (1½). Plant root hair cells are long to absorb water (1).	1+1½+1½+1
	OR: Slurry of cow dung and water is fed in a digester (1). Microbes break it down without oxygen (1). Methane-rich gas collects (1). Advantage: clean fuel (1). Advantage: leftover slurry is used as manure (1).	1+1+1+1+1
25	Diagram (1). Zinc container negative (1). Carbon rod with cap positive (1). Paste electrolyte (1). Chemical reaction produces current and cell is single-use (1).	1+1+1+1+1

	OR: Insert copper and iron into lemons (1). Join several lemons and connect LED (1). LED glows (1). Metals are electrodes, juice is electrolyte (1). Chemical reaction produces current (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** Which colour is the nicest?	Opinions cannot be measured
2	**(c)** curiosity taking flight	Curiosity soars
3	**(a)** unit	Numbers need units
4	**(a)** Yeast	Used in baking
5	**(b)** plants and animals	Complex bodies
6	**(c)** Lactobacillus	A bacterium
7	**(a)** mould	Stopped bacterial growth on a petri dish
8	**(c)** misuse of antibiotics	Incomplete or unnecessary doses
9	**(b)** contaminated water and food	Intestinal bacterial infection
10	**(d)** Alessandro Volta	Metals and liquid give current
11	**(c)** gets deflected	Current produces a magnetic field
12	**(a)** positive terminal	LED works in one direction only

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	They record every observation (1). They help in explaining results and planning new tests (1).	1+1
14	Cell, tissue, organ (1). Organ system, organism (1).	1+1
15	Through air by coughing or sneezing (1). Through contaminated food and water or vectors (1).	1+1
16	Health is a state of complete physical, mental and social wellbeing (1). Not merely the absence of disease (1).	1+1
17	By reversing the direction of current (1). Poles of the coil then interchange (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Dough thickness (1). Type of flour (1). Oil temperature or way of dropping dough (1).	1+1+1
19	Neat diagram (1). Labels: cell membrane, cytoplasm, nucleus (2).	1+2
20	Cause: bacteria (1). Site: intestine with diarrhoea and dehydration (1). Prevention: boiled water, hygiene, vaccination (1).	1+1+1
21	Nichrome becomes hotter (1). It has higher resistance (1). More electrical energy changes to heat (1).	1+1+1
22	Curiosity (1). Careful observation (1). Willingness to test and change ideas using evidence (1).	1+1+1
	OR: (i) Oil temperature (1). (ii) Time to puff or whether it puffs (1). (iii) Thickness, size, flour, way of dropping (1).	1+1+1
23	Tissue: group of similar cells, e.g. muscle tissue (1½). Organ: tissues working together, e.g. heart (1½).	1½+1½
	OR: Sugar solution with yeast in a test tube, balloon on mouth (1). Kept warm for 3–4 hours (1). Balloon inflates due to carbon dioxide from fermentation (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Records preserve evidence (1). They allow comparison of trials (1). Include dough size and thickness (1). Include oil temperature and time (1). Include sensory notes such as splatter or smoke (1).	1+1+1+1+1
	OR: (i) How does oil temperature affect puffing? (1). (ii) Oil temperature at three levels (1). (iii) Same dough, thickness, size, flour, dropping method (1). (iv) Whether it puffs and time to puff (1). (v) Table of trials with notes on splatter, smell and smoke (1).	1+1+1+1+1
25	Milkmaids with cowpox did not get smallpox (1). Hypothesis: cowpox material protects (1). Experiment on a boy (1). Result: resistance to smallpox (1). Mass vaccination eradicated smallpox (1).	1+1+1+1+1

	OR: Immune system recognises pathogens (1). Vaccine shows a harmless form (1). Memory formed (1). Weakened or dead pathogens, or parts of pathogen (1). Example: tetanus shot (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

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SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** a focused question	Questions are the starting point
2	**(c)** observation, measurement and experiment	Scientific method
3	**(c)** starting an investigation	Curiosity-driven question
4	**(b)** microalga used as food supplement	Rich in protein
5	**(b)** cell	All living bodies are made of cells
6	**(c)** cell → tissue → organ → organ system → organism	Small to large
7	**(c)** cancer	Biomedical researcher
8	**(a)** Drinking boiled water	Boiling kills pathogens
9	**(d)** Dengue	Mosquito-borne viral disease
10	**(a)** more cells are connected	More current
11	**(b)** it becomes hot	Safety first
12	**(a)** Hans Christian Oersted	In 1820

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Stopwatch measures time (1). Thermometer measures oil temperature (1).	1+1
14	Bacteria (1). Amoeba or yeast (1).	1+1
15	Symptom is what the patient feels, e.g. pain (1). Sign is seen or measured, e.g. fever or rash (1).	1+1
16	Electric heater (1). Electric iron or kettle (1).	1+1
17	Lifts iron and steel objects in factories (1). Releases them when current is switched off (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Everyday events like dough rising can be investigated (1½). No fancy laboratory is needed to begin (1½).	1½+1½
19	Very small, not like other microorganisms (1). Reproduce only inside a host cell (1). Can cause diseases (1).	1+1+1
20	Processed and sugary food (1). Less exercise (1). People live longer (1).	1+1+1
21	Insert copper wire and iron nail in each lemon (1). Join several lemons in series and connect an LED (1). LED glows; reverse connections if it does not (1).	1+1+1
22	They are very small and light (1). They need little food (1). They can live in air, water, soil, plants and animals (1).	1+1+1
	OR: Rhizobium lives in root nodules (1). It fixes nitrogen from air into usable form (1). The plant gets nitrogen and bacteria get shelter and food (1).	1+1+1
23	They kill bacteria by targeting parts that differ from human cells (1½). Viruses lack these parts so antibiotics do not act (1½).	1½+1½
	OR: Antibiotics are used unnecessarily or in incomplete doses (1). Some bacteria survive (1). Survivors multiply and pass on resistance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Both tubes have sugar solution (1). Yeast only in B, balloons on both, kept warm (1). Balloon of B inflates, A does not (1). Yeast produced a gas (1). Gas is carbon dioxide by fermentation (1).	1+1+1+1+1
	OR: Cell wall present in plant and bacteria (1). Chloroplast only in plant (1). Nucleus well-defined in plant and animal (1). Nucleoid in bacteria (1). Size, vacuole or other point (1).	1+1+1+1+1

25	Galvani noticed frog's leg kicking with copper and iron (1½). He believed electricity came from the frog (1). Volta believed it came from metals (1). He used saltwater-soaked paper and still got current (1). Led to the first battery (½).	1½+1+1+1+½
	OR: Compass deflects near current wire (1). Current produces magnetic field (1). Coil on iron core forms electromagnet (1). Three factors: current, turns, iron core (2).	1+1+1+2

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